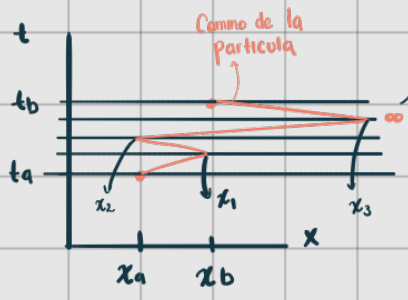


Un camino puede ir a infinito y luego volver

↦ Subdividimos el tiempo en N partes iguales de tamaño ε

$\varepsilon = \frac{t_b - t_a}{N}$

↦ La distancia entre x_a y x_b pueden ir de $-\infty$ a ∞

$$\int_a^b f(x)\,dx \approx \varepsilon \sum_{n=0}^{N} f(x_n)$$

$$\int_{t_a}^{t_b} \tfrac{1}{2} m v^2 - V(x)\,dt \approx \varepsilon \sum_{n=0}^{N-1} \tfrac{1}{2} m v_n^2 - V(x_n)$$

$$v_n = \frac{x_{n+1} - x_n}{\varepsilon}$$

$$v_N = \frac{x_{N+1} - x_N}{\varepsilon}$$

Solo hay N partes no N+1, luego la suma va hasta N-1

$$S \approx \sum_{n=0}^{N-1} \left[\tfrac{1}{2} m \varepsilon \frac{(x_{n+1} - x_n)^2}{\varepsilon^2} - \varepsilon V(x_n) \right]$$

$$S \approx \sum_{n=0}^{N-1} \left[\frac{m(x_{n+1} - x_n)^2}{2\varepsilon} - \varepsilon V(x_n) \right]$$

Amplitud de probabilidad de uno de los caminos

$$\phi_j = A \exp\left\{ \tfrac{i}{\hbar} \sum_{n=0}^{N-1} \left[\frac{m(x_{n+1} - x_n)^2}{2\varepsilon} - \varepsilon V(x_n) \right] \right\}$$

$\phi = \Sigma \phi_j$ → sumar todas las amplitudes

$$\phi = \underbrace{\phi_{x_1, x_2 \dots x_{N-1}}}_{\text{1 camino}} + \underbrace{\phi_{x_1, x_2 \dots x_{N-1}}}_{\text{otro camino}} \dots$$

$$K(b,a) = A \int_{-\infty}^{\infty} dx_1 \int_{-\infty}^{\infty} dx_2 \cdots \int_{-\infty}^{\infty} dx_{N-1}\; e^{\frac{i}{\hbar} \sum_{n}^{N-1} \frac{m(x_{n+1} - x_n)^2}{2\varepsilon} - \varepsilon V(x_n)} = \int \mathcal{D}x\, e^{\frac{i}{\hbar} S}$$

↳ Integral de camino de Feynman

No incluimos x_0 y x_N ya que son fijos

$$K(b,a) = A \int_{-\infty}^{\infty} dx_1 \int_{-\infty}^{\infty} dx_2 \cdots \int_{-\infty}^{\infty} dx_{N-1}\; e^{\frac{i}{\hbar} \sum_{n}^{N-1} \frac{m(x_{n+1} - x_n)^2}{2\varepsilon} - \varepsilon V(x_n)} = \int \mathcal{D}x\, e^{\frac{i}{\hbar} S}$$

Integrales Gaussianas

$$\int_{-\infty}^{\infty} dy_1 \int_{-\infty}^{\infty} dy_2 \cdots \int_{-\infty}^{\infty} dy_{N-1}\, e^{-b(a_1 y_1^2 + a_2 y_2^2 \cdots a_{n-1} y_{n-1}^2)} = \sqrt{\frac{\pi^{N-1}}{b^{N-1}\, a_1 \cdot a_2 \cdots a_{N-1}}}$$

↦ $\int_{-\infty}^{\infty} dy_1\, e^{-b a_1 y_1^2} = \sqrt{\frac{\pi}{b a_1}}$ Integral de gauss

↦ $a_1 y_1^2 + a_2 y_2^2 = 3y_1^2 + 1y_2^2 \rightarrow \underbrace{\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}}_{\text{Base Canonica}} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \underbrace{\begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}}_{\text{Base B}} \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Cambio de base $y_1, y_2 \rightarrow z_1, z_2$ $\quad \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \underbrace{\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}}_{M} \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$

Ejemplo $\int_{-\infty}^{\infty} dz_1 \int_{-\infty}^{\infty} dz_2\, e^{-b(2z_1^2 - 2z_1 z_2 + 2z_2^2)} = \sqrt{\frac{\pi^2}{b^2 \det(M)}}$

$\det(M) = N$

↳ Solo particula libre

$$\int_{-\infty}^{\infty} dz_1 \int_{-\infty}^{\infty} dz_2 \cdots \int_{-\infty}^{\infty} dz_{N-1}\, e^{-b\left(z_1^2 + (z_2 - z_1)^2 + (z_3 - z_2)^2 \cdots z_{N-1}^2\right)} = \sqrt{\frac{\pi^{N-1}}{b^{N-1} \det M}} = \sqrt{\frac{\pi^{N-1}}{b^{N-1} N}}$$

$$\int_{-\infty}^{\infty} dz_1 \int_{-\infty}^{\infty} dz_2 \cdots \int_{-\infty}^{\infty} dz_{N-1}\, e^{-b\, \vec{z}^T \cdot M \cdot \vec{z}} = \sqrt{\frac{\pi^{N-1}}{b^{N-1} \det M}}$$

general

Expansion de la Accion a segundo Orden

Usando series de Taylor

$$S[x(t)] = \int_{t_a}^{t_b} L(x,\dot{x})\,dt$$

$$S[x+z] \simeq \int_{t_a}^{t_b} \left\{ L(x,\dot{x}) - z\,V'(x) + m\dot{x}\dot{z} - \tfrac{1}{2}V''(x)z^2 + \tfrac{1}{2}m\dot{z}^2 \right\} dt$$

$$\boxed{S[x+z] \simeq S^{cl} + \int_{t_a}^{t_b} \left(\tfrac{1}{2}m\dot{z}^2 - \frac{V''(x)}{2}z^2\right) dt} \qquad z(t_a) = z(t_b)$$

$$\boxed{S^{cl} = \tfrac{1}{2}\,m\,\frac{(x_b-x_a)^2}{t_b-t_a}}$$

Propagador de una particula libre

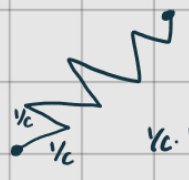
$$K(a,b) = A\int_{-\infty}^{\infty} dz_1 \int_{-\infty}^{\infty} dz_2 \cdots \int_{-\infty}^{\infty} dz_{N-1}\; e^{i/\hbar\left(S^{cl} + \varepsilon\sum_{n=0}^{N-1}\frac{1}{2}m\frac{(z_{n+1}-z_n)^2}{\varepsilon^2}\right)}$$

$$K(a,b) = Ae^{i/\hbar S^{cl}} \int_{-\infty}^{\infty} dz_1 \cdots \int_{-\infty}^{\infty} dz_{N-1}\; e^{\frac{-m}{2i\hbar\varepsilon}\left[z_1^2 + (z_2-z_1)^2 \cdots z_{N+1}^2\right]}$$

$\hookrightarrow -b$

$$K(a,b) = Ae^{i/\hbar S^{cl}} \sqrt{\frac{\pi^{N-1}}{\left(\frac{m}{2i\hbar\varepsilon}\right)^{N-1}\cdot N}}$$

$$K(a,b) = A\sqrt{\left(\frac{2\pi\hbar i\varepsilon}{m}\right)^{N-1}\cdot\frac{1}{N}}\; e^{i/\hbar\frac{m}{2}\frac{(x_b-x_a)^2}{t_b-t_a}}$$



$1/c \cdot 1/c \cdots 1/c^N = A$

$$K(a,b) = 1/c^N\sqrt{\left(\frac{2\pi\hbar i\varepsilon}{m}\right)^{N-1}\cdot\frac{1}{N}}\; e^{i/\hbar\frac{m}{2}\frac{(x_b-x_a)^2}{t_b-t_a}} \qquad c = \sqrt{\frac{2\pi\hbar i\varepsilon}{m}}$$

$$K(a,b) = \frac{1}{\left(\frac{2\pi\hbar i\varepsilon}{m}\right)^{N/2}} \left(\frac{2\pi\hbar i\varepsilon}{m}\right)^{N/2}\left(\frac{2\pi\hbar i\varepsilon}{m}\right)^{-1/2} N^{-1/2}\; e^{i/\hbar\frac{m}{2}\frac{(x_b-x_a)^2}{t_b-t_a}}$$

$$\varepsilon = \frac{t_b-t_a}{N}$$

$$\boxed{K(x_b,t_b,x_a,t_a) = \sqrt{\frac{m}{2\pi i\hbar(t_b-t_a)}}\; e^{\frac{im}{2\hbar}\frac{(x_b-x_a)^2}{t_b-t_a}}}$$

Amplitud de probabilidad de pasar de $x_a \to x_b$

Proyecto:

Motivación: Calcularemos E_0 usando la función de Partición Z, con un tiempo imaginario, ya que eso suprime las energías mayores

Normalmente Operador evolución temporal $e^{\frac{-i}{\hbar}\hat{H}t}$

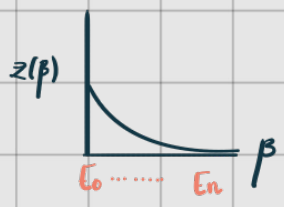
Rotación W. $t = -i\tau$, $\beta = \tau/\hbar$ $\quad e^{\frac{-\hat{H}\tau}{\hbar}} \rightarrow e^{-\beta\hat{H}}$

$$Z(\beta) = Tr\left(e^{-\beta\hat{H}}\right) \qquad e^{-\beta\hat{H}}|n\rangle = e^{-\beta E_n}|n\rangle \qquad \langle n|n\rangle = 1$$

$$Z(\beta) = \sum \langle n|e^{-\beta\hat{H}}|n\rangle$$

$$Z(\beta) = \sum \langle n|e^{-\beta E_n}|n\rangle$$

$$Z(\beta) = \sum e^{-\beta E_n}$$



Ya no es oscilatorio → Decae (fases complejas)

$$Z(\beta) \simeq e^{-\beta E_0}$$

$$\downarrow$$

$$E_0 \approx -\frac{\ln Z(\beta)}{\beta}$$ → El reto es calcular $Z(\beta)$ numericamente

Como E_n es cuantizada $\quad Z(\beta) = Tr\left(e^{-\beta\hat{H}}\right) = \sum \langle n|e^{-\beta\hat{H}}|n\rangle$

Como x es continua $\quad Z(\beta) = Tr\left(e^{-\beta\hat{H}}\right) = \int dx \langle x|e^{-\beta\hat{H}}|x\rangle$

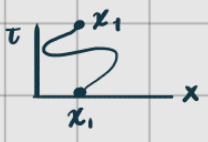
Importante!!! El elemento de matriz comienza y termina en la misma posición x. Es decir trayectorias cerradas en la representación de integral de camino. $x(\beta) = x(0)$

$Z(\beta) =$ 

Traza impone esa condición

$x_1 \rightarrow x_1 \quad a_{11}$
$x_2 \rightarrow x_2 \quad a_{22}$
$x_3 \rightarrow x_3 \quad a_{33}$

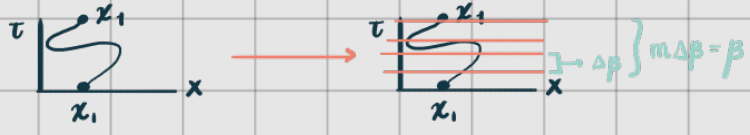
$$Z(\beta) = Tr(e^{-\beta\hat{H}}) = \int dx_1 \langle x_1 | e^{-\beta\hat{H}} | x_1 \rangle$$ → Describe una trayectoria continua



$$Z(\beta) = \int dx_1 \langle x_1 | e^{-\beta(\hat{K}+\hat{V})} | x_1 \rangle$$

Surge un problema $[\hat{K},\hat{V}] \neq 0$ $\quad e^{-\beta(\hat{K}+\hat{V})} \neq e^{-\beta\hat{K}}\, e^{-\beta\hat{V}}$

Solución Dividir el parametro β en m segmentos $\quad \Delta\beta = \beta/m$ (Dividir el tiempo)



↦ Describe una trayectoria segmentada

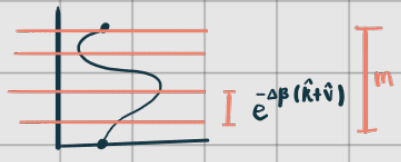
Para aproximar $e^{-\beta(\hat{K}+\hat{V})} \approx e^{-\beta\hat{K}}\, e^{-\beta\hat{V}}$ usamos series de Taylor (Aproximación de 1er orden de Trotter)

$$\left.\begin{aligned} e^{\Delta t A} &\approx I + \Delta t\, A \\ e^{\Delta t B} &\approx I + \Delta t\, B \end{aligned}\right\} e^{\Delta t A}\, e^{\Delta t B} \approx I + \Delta t (A+B) + \cdots$$

Por otro lado $e^{\Delta t(A+B)} \approx I + \Delta t (A+B) \cdots \approx e^{\Delta t A}\, e^{\Delta t B}$ (Solo funciona para intervalos muy pequeños)

Con este truco pasamos de $e^{-\beta(\hat{K}+\hat{V})} \neq e^{-\beta\hat{K}}\, e^{-\beta\hat{V}}$ a que para intervalos muy pequeños

$$e^{-\Delta\beta(\hat{K}+\hat{V})} \approx e^{-\Delta\beta\hat{K}}\, e^{-\Delta\beta\hat{V}}$$



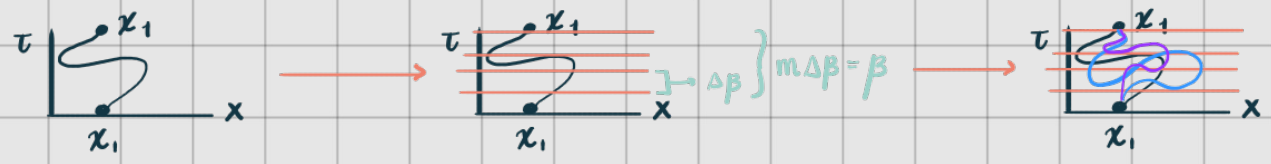
Llamemos $\hat{U} \approx e^{-\Delta\beta\hat{K}}\, e^{-\Delta\beta\hat{V}}$ ahora si sustituyendo en la traza

$$Z(\beta) = \int dx_1 \langle x_1 | e^{-\beta(\hat{K}+\hat{V})} | x_1 \rangle$$

$$Z(\beta) = \int dx_1 \langle x_1 | \left(e^{-\Delta\beta\hat{K}}\, e^{-\Delta\beta\hat{V}}\right)\left(e^{-\Delta\beta\hat{K}}\, e^{-\Delta\beta\hat{V}}\right) \cdots | x_1 \rangle$$ → m segmentos

$$Z(\beta) = \int dx_1 \langle x_1 | \hat{U}_1 \hat{U}_2 \hat{U}_3 \cdots \hat{U}_m | x_1 \rangle$$

Ahora introducimos **Infinitas trayectorias segmentadas**



Nos aprovechamos de $\hat{I} = \int dx\, |x\rangle\langle x|$

Ejemplo $m=3$ $\quad Z_3 = \int dx_1 \langle x_1|\hat{U}\cdot\hat{U}\cdot\hat{U}|x_1\rangle$

$$Z_3 = \int dx_1 \langle x_1|\hat{U}\left(\int dx_3\, |x_3\rangle\langle x_3|\right)\hat{U}\left(\int dx_2\, |x_2\rangle\langle x_2|\right)\hat{U}|x_1\rangle$$

$$Z_3 = \int dx_1 \int dx_2 \int dx_3 \; \langle x_1|\hat{U}|x_3\rangle\langle x_3|\hat{U}|x_2\rangle\langle x_2|\hat{U}|x_1\rangle$$

Generalizado $\quad Z_m = \int dx_1 \int dx_2 \cdots \int dx_m \prod_{j=1}^{m} \langle x_{j+1}|\hat{U}|x_j\rangle$ $\quad$ Recordemos $x_{m+1} = x_1$

$$\langle x_{j+1}|\hat{U}|x_j\rangle = \langle x_{j+1}|e^{-\Delta\beta\hat{K}}e^{-\Delta\beta\hat{V}}|x_j\rangle = e^{-\Delta\beta V(x_j)}\underbrace{\langle x_{j+1}|e^{-\beta\hat{K}}|x_j\rangle}$$

Propagador de una partícula libre

$$\langle x_{j+1}|\hat{U}|x_j\rangle = e^{-\beta V(x_j)}\sqrt{\frac{\mu}{2\pi\hbar^2\Delta\beta}}\; e^{\frac{-\mu(x_{j+1}-x_j)^2}{2\hbar^2\Delta\beta}}$$

$$= \sqrt{\frac{\mu}{2\pi\hbar^2\Delta\beta}}\; \exp\left[\frac{-\mu(x_{j+1}-x_j)^2}{2\hbar^2\Delta\beta} - \beta V(x_j)\right]$$

Finalmente

$$Z_m = \int dx_1 \int dx_2 \cdots \int dx_m \prod_{j=1}^{m}\left(\langle x_{j+1}|\hat{U}|x_j\rangle\right) \qquad \text{Como } \prod_{j=1} e^j = \sum_{j=1} e^j$$

$$Z_m = \int dx_1 \int dx_2 \cdots \int dx_m \prod_{j=1}^{m}\left(\sqrt{\frac{\mu}{2\pi\hbar^2\Delta\beta}}\; \exp\left[\frac{-\mu(x_{j+1}-x_j)^2}{2\hbar^2\Delta\beta} - \beta V(x_j)\right]\right)$$

$$Z_m = \left(\frac{\mu}{2\pi\hbar^2\Delta\beta}\right)^{m/2} \int dx_1 \int dx_2 \cdots \int dx_m \; \exp\left(-\sum_{j=1}^{m}\left[\frac{\mu(x_{j+1}-x_j)^2}{2\hbar^2\Delta\beta} + \beta V(x_j)\right]\right)$$

$$Z_m = \left(\frac{\mu}{2\pi\hbar^2\Delta\beta}\right)^{m/2} \int dx_1 \int dx_2 \cdots \int dx_m \; \exp\left(-\sum_{j=1}^{m}\left[\frac{\mu(x_{j+1}-x_j)^2}{2\hbar^2\Delta\beta} + \beta V(x_j)\right]\right)$$

Si recordamos $\Delta\tau = \hbar\Delta\beta$, podemos expresar el exponente como una acción

$$S_m = \sum_{j=1}^{m}\left[\frac{\mu(x_{j+1}-x)^2}{2\Delta\tau} + \Delta\tau V(x_j)\right]$$

$$Z_m = \left(\frac{\mu}{2\pi\hbar^2\Delta\beta}\right)^{m/2} \int dx_1 \cdots \int dx_m \; e^{-S_m/\hbar} \longrightarrow$$ Tiene pinta de una integral de Camino de Feynman

Otra forma mas rapida

$$\hat{H} = \hat{p}^2/2\mu - \kappa/r$$
$$\hat{H} = \hat{p}^2/2\mu - e^2/4\pi\varepsilon_0 r$$

$$S^{cl}(r) = \int_{t_i}^{t_f} dt \left(\frac{\mu}{2}\left(\frac{dr}{dt}\right)^2 + \frac{\kappa}{r(t)}\right)$$

$$K(r_f, t_f; r_i, t_i) = \int_{r_i}^{r_f} \mathcal{D}r(t)\; e^{\frac{i}{\hbar}S[r]}$$

Cambio $t = -i\tau$ $\quad dt = -i\,d\tau$ $\quad \frac{dr}{dt} = \frac{dr}{d\tau}\frac{d\tau}{dt} = i\frac{dr}{d\tau}$ $\quad \left(\frac{dr}{dt}\right)^2 = -\left(\frac{dr}{d\tau}\right)^2$

$$S = \int dt \left(\frac{\mu}{2}\left(\frac{dr}{dt}\right)^2 + \frac{\kappa}{r}\right)$$
$$S = i\int d\tau \left(\frac{\mu}{2}\left(\frac{dr}{d\tau}\right)^2 - \frac{\kappa}{r}\right)$$

Acción Euclidea $S_E = \int_{\tau_i}^{\tau_f} d\tau \left(\frac{\mu}{2}\left(\frac{dr}{d\tau}\right)^2 - \frac{\kappa}{r(\tau)}\right)$ $\qquad S = iS_E$

$$K(r_f, t_f; r_i, t_i) = \int_{r_i}^{r_f} \mathcal{D}r(t)\; e^{\frac{i}{\hbar}S[r]} \longrightarrow K(r_f, \tau_f; r_i, \tau_i) = \int_{r_i}^{r_f} \mathcal{D}r(\tau)\; \underbrace{e^{-\frac{S_E}{\hbar}}}$$

Peso estadistico

Monte Carlo (Tenemos que discretizar)

$$\Delta\tau = \frac{\tau_f - \tau_i}{m} \qquad r_0, r_1 \cdots r_m \qquad \frac{dr}{d\tau} \approx \frac{r_{j+1} - r_j}{d\tau}$$

$$S_E = \int_{\tau_i}^{\tau_f} d\tau \left(\frac{M}{2}\left(\frac{dr}{d\tau}\right)^2 - \frac{K}{r(\tau)} \right) \longrightarrow S_{E,m} = \sum_{j=1}^{m} \left[\frac{M(r_{j+1} - r_j)^2}{2\Delta\tau} - \frac{\Delta\tau K}{r_j} \right]$$

$e^{-S_E/\hbar}$ (Peso estadístico w)

$$w_{pot}(r_j) = e^{\left(\frac{\Delta\tau K}{\hbar r_j}\right)} \quad \text{[adimensional]}$$

$$K(r_f, t_f; r_i, t_i) = \int_{r_i}^{r_f} \mathcal{D}r(t)\, e^{\frac{i}{\hbar} S(r)} \longrightarrow K(r_f, \tau_f; r_i, \tau_i) = \int_{r_i}^{r_f} \mathcal{D}r(\tau)\, \underbrace{e^{-\frac{S_E}{\hbar}}}_{\text{Peso estadístico}}$$

$$\int_{r_i}^{r_f} \mathcal{D}r(\tau)\, \underbrace{e^{-\frac{\sum_{j=1}^{m}\left[\frac{M(r_{j+1}-r_j)^2}{2\Delta\tau} - \frac{\Delta\tau K}{r_j}\right]}{\hbar}}}_{\text{Peso estadístico}}$$

Surge el problema del peso del potencial

$$w_{pot} = e^{-\frac{\Delta\tau V(r)}{\hbar}} = e^{\frac{\Delta\tau K}{\hbar r}} \qquad \lim_{r \to 0^+} e^{\frac{\Delta\tau K}{\hbar r}} = +\infty$$

el peso primitivo es no acotado

Miremos su integral

Contribución radial del peso primitivo

$$I(\varepsilon) = \int e^{\Delta\tau K/\hbar r}\, dV$$

$$I(\varepsilon) = \int e^{\Delta\tau K/\hbar r}\, 4\pi r^2 dr$$

r^2 no compensa $e^{a/r}$ → diverge

$$I(\varepsilon) = 4\pi \int_0^{\varepsilon} r^2 e^{\Delta\tau/\hbar r}\, dr \to +\infty$$

1) $e^{a/r} \to 0$

2) Integral con $4\pi r^2 dr$ tambien diverge

3) Suma diverge

4) El termino cinetico no compensa la divergencia tampoco

Solucion $Z_m = \left(\frac{M}{2\pi\hbar^2\Delta\beta}\right)^{m/2} \int dx_1 \cdots \int dx_m \, e^{-\sum_{j=0}^{m}\left[\frac{M(x_{j+1}-x_j)^2}{2\Delta\tau} + \Delta\tau V(x_j)\right]/\hbar}$ → Tiene pinta de una integral de camino de Feynman

$$K = \int_{r_0}^{r_f} \mathcal{D}r(\tau)\, e^{-\sum_{j=0}^{m}\left[\frac{M(r_{j+1}-r_j)^2}{2\Delta\tau} - \frac{\Delta\tau K}{r_j}\right]/\hbar}$$

Peso estadistico

$\Delta\beta = \Delta\tau = \beta/m$

Solucion en general

$\langle x | e^{-\tau(\hat{K}+\hat{V})} | x' \rangle$ $x \neq x'$ elemento de matriz x_{i+1}, x_i Intermedias

$$Z = Tr\left(e^{-\tau(\hat{K}+\hat{V})}\right) = \int dx \, \langle x | e^{-\tau(\hat{K}+\hat{V})} | x \rangle$$

$$K(x,t,x_0,t_0) = \langle x | e^{-i/\hbar K t} | x' \rangle = \int_{x'}^{x} \mathcal{D}x \, e^{-S/\hbar}$$